Task-1) List down 5 types of problem commonly seen in nature which can be solved by supervised and unsupervised learning each. Mention which model will work best, what will be the benefits etc.

First categorisation of each example starts with

If output is a number – Regression

- Linear / multiple
  PROS -> fast
  Cons - > Assume straight line relationship , sensitive to outliers

If output is a category – Classification

- Logistic regression
  PROS -> Probability output ,
  CONS -> Not for complex patterns
- Decision tree
  PROS -> easy , simple , can tackle non linear data
  CONS -> can overfit easily , and can change drastically for small change in data
- Random forest
  PROS -> less overfitting , better with complex data , Handles noise , high accuracy
  CONS -> Harder to interpret

If data has no labels and you want groups – Clustering

- K-means
  PROS -> fast , simple
  CONS -> k needs to be fixed beforehand , sensitive to outliers
- PCA(Principle component analysis)(reduces number of features)
  PROS -> reduce complexity , speeds up training
  CONS -> Some information is lost
- SVD(Singular value decomposition) (breaks large matrix into smaller ones)
  PROS -> Great with recommendation , works well with smaller data also
  CONS -> more mathematical , expensive

Supervised Learning Examples

1. **Disease Diagnosis**
   Problem: Predicting whether a patient has cancer or diabetes based on symptoms and test results.
   Type: Classification
   Best Model: Random Forest
   Why: Output is a category — disease or no disease. Random Forest handles noisy medical data better than a single decision tree because it combines multiple trees and takes a majority vote.
   Benefit: High accuracy, works well with high-dimensional medical data, reduces misdiagnosis

2. **House Price Prediction**

Problem: Predicting the price of a house based on features such as area, number of bedrooms, location, and age of the house.
Type: Regression
Best Model: Multiple Linear Regression
Why: House price is a continuous numerical value. Multiple Linear Regression can model the relationship between several input variables and the house price.
Benefit: Helps buyers, sellers, and real estate companies estimate property values quickly and consistently.

3. **Credit Risk Scoring**
Problem: Predicting whether a loan applicant will default or repay.
Type: Classification
Best Model: Logistic Regression
Why: Output is binary — default or repay. Logistic Regression gives a probability score (e.g. 78% chance of default) which banks use to set approval thresholds.
Benefit: Reduces bad loans, automates decisions, saves manual review time
4. **Stock Price Prediction**
Problem: Predicting what tomorrow's stock price will be based on historical data.
Type: Regression
Best Model: Linear Regression / Polynomial Regression
Why: Output is a continuous number — the price. Linear regression for simple trends, Polynomial for curved patterns.
Benefit: Helps investors make data-driven decisions, reduces emotional trading.
5. **Image Recognition**
Problem: Identifying what object is present in an image.
Type: Classification
Best Model: Decision Tree / Random Forest
Why: Multi-class output — cat, dog, car etc. From the course diagram, Decision Tree and Random Forest are the available classifiers for this type of problem.
Benefit: Powers medical imaging, security cameras, manufacturing defect detection.

Unsupervised learning examples

1. **Cancer Cell Detection in Medical Images**
Problem: Finding tumour regions in MRI scans without any pre-labeled data.
Type: Clustering
Best Model: K-Means
Why: No pre-labeled tumour regions exist. K-Means groups pixels by similarity — abnormal tissue clusters separately from healthy tissue automatically.
Benefit: Helps doctors identify regions of interest faster, improves early diagnosis accuracy.

2. **Climate Anomaly Detection via Satellite**
Problem: Detecting vegetation changes or climate disasters from satellite images.
Type: Clustering
Best Model: K-Means + PCA
Why: Satellite images have thousands of dimensions. PCA compresses them first, then K-Means clusters normal vs anomalous vegetation patterns.
Benefit: Early disaster warning system, no need for manually labeled satellite data.
3. **Customer Segmentation in Retail**
Problem: Grouping customers by purchasing behaviour without knowing categories

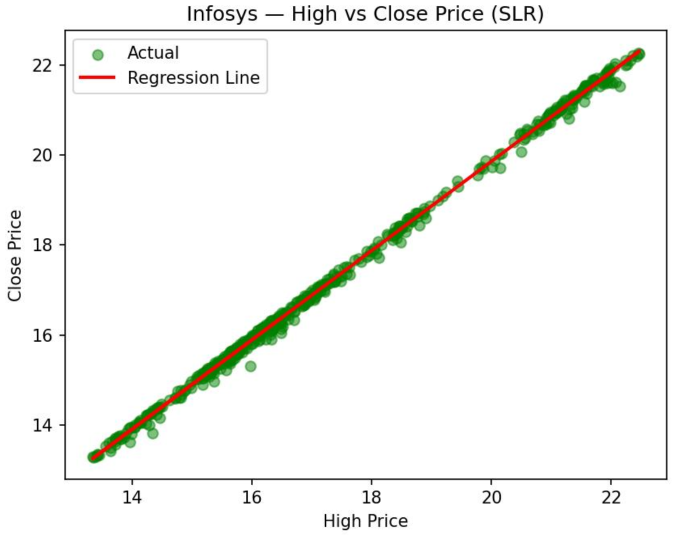
in advance.
Type: Clustering
Best Model: K-Means
Why: No predefined customer categories exist. K-Means finds natural groups like high spenders, seasonal buyers, and bargain hunters from raw purchase data.
Benefit: Enables targeted marketing, personalized offers, better product recommendations.

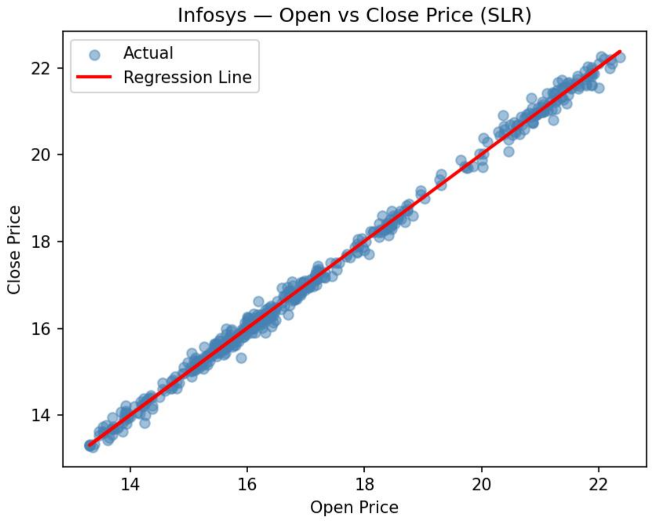
4. **Netflix Content Recommendation**
   Problem: Suggesting content to users based on watch history without predefined preference labels.
   Type: Clustering
   Best Model: SVD (Singular Value Decomposition)
   Why: SVD breaks down the user-movie matrix into hidden factors such as "likes action" or "watches late night" and finds patterns no human would label manually.
   Benefit: Enhances user experience, increases watch time, reduces subscriber churn.
5. **Fraud Detection in Banking**
   Problem: Detecting unusual transaction patterns without knowing all fraud types in advance.
   Type: Clustering
   Best Model: K-Means + PCA
   Why: Normal transactions cluster tightly together. Fraudulent ones don't fit any cluster and appear as outliers far from all centroids. PCA helps visualise this in 2D.
   Benefit: Catches new fraud types that rule-based systems miss, adapts to unknown attack patterns automatically.

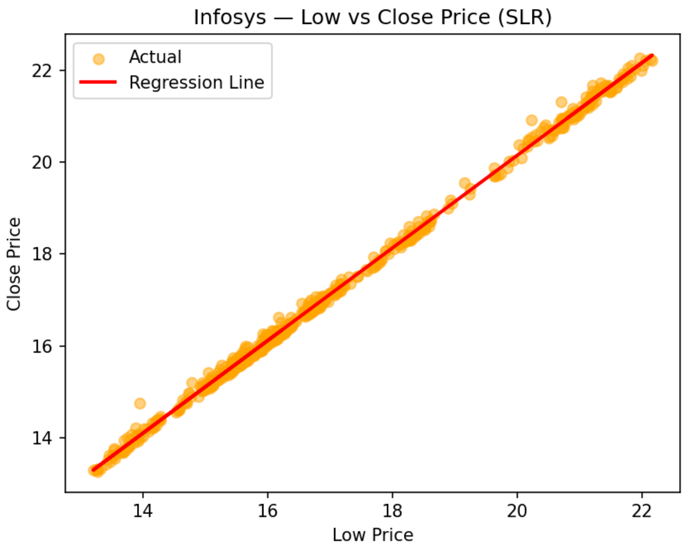
Task-2) Identify stock values where we can apply regression and apply linear regression to them. Is SLR fine to predict stock value if not what should be feature space for MLR.

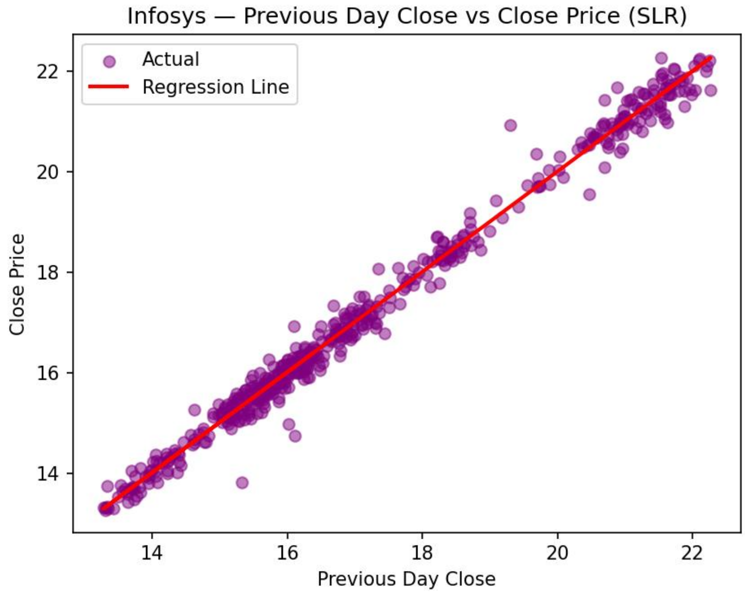
For this task I chose Infosys (INFY) as my stock. I experimented with different input-output combinations to see how SLR behaves on stock data:

- Opening Price → Closing Price
- High Price → Closing Price
- Low Price → Closing Price
- Previous Day Close → Next Day Close



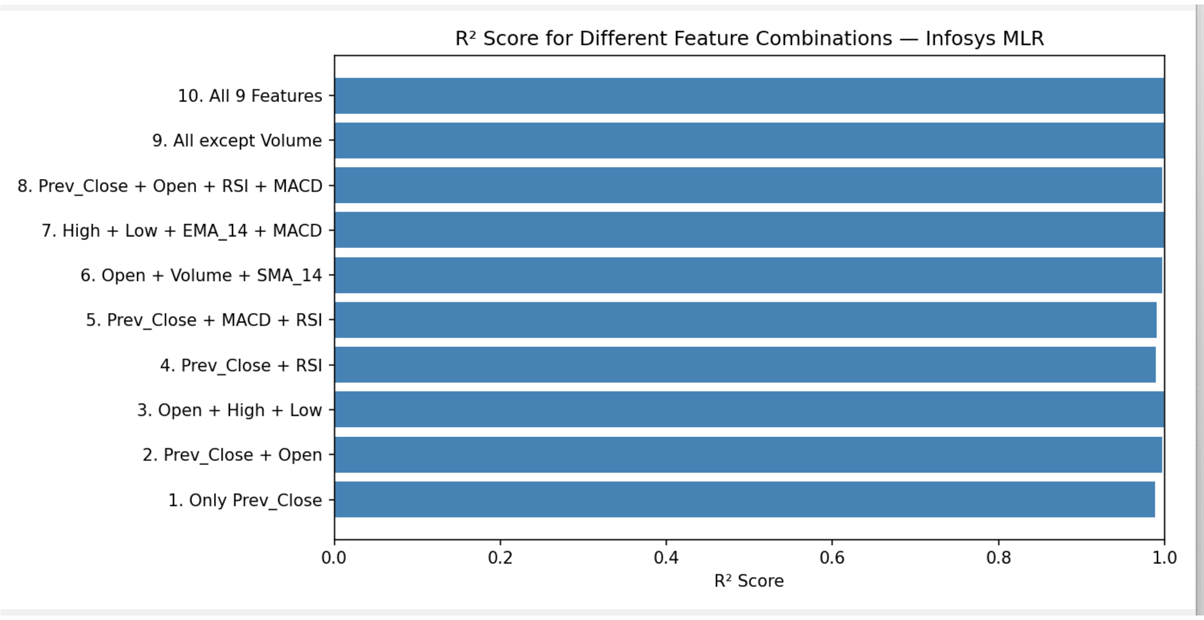






- Among all four combinations tested on Infosys stock data, High Price vs Close Price showed the strongest linear relationship, followed closely by Open Price vs Close Price. Low Price vs Close Price also performed well. Previous Day Close vs Today's Close had the most scatter, showing that yesterday's price alone is not a reliable predictor of today's close.
- However, even the best performing combination (High vs Close) is not practically useful for real prediction — because you only know the High Price after the trading day ends, at which point the Close Price is already known too.
- This proves that SLR is not sufficient for stock price prediction. A proper MLR model with features known before market close — such as Opening Price, Trading Volume, Moving Averages, PE Ratio and Market Sentiment — would be needed for any real prediction system.

  Now using MLR the feature space should be :



$R^2 = 1.0$ exactly
Bad – overfitting or multicollinearity
R^2 slightly below 1.0
Good — real
R^2 much below 1.0
Bad — underfitting, too few features

- These are some of the random combinations for MLR that I tried and several were overfitted like 10th which contained all the features
- The 1st one is the underfitting one as you can see the value of R^2 is less as it is nothing but SLR
- Also the 3rd one shows multicollinearity as Open, High, Low and Close all move together on the same day. Which makes features related to each other

So while choosing the features you need to take care that u don't overfit or underfit neither multicollinearity is there